

Financial Health Score Analysis

1. Introduction

Financial stability plays an important role in personal financial management. However, many individuals struggle to maintain a healthy financial balance due to high spending habits and debt obligations.

FinTech companies often analyze customer financial behavior to understand spending patterns, savings habits, and debt burden in order to recommend financial products or identify financially vulnerable customers.

This project analyzes financial behavior by developing a Financial Health Score model based on key financial indicators.

2. Objectives

The main objectives of this project are:

- Analyze financial behavior of users based on income, expenses, savings, and debt payments.
- Develop financial metrics such as savings ratio, spending ratio, and debt burden.
- Create a Financial Health Score to evaluate financial stability.
- Classify users into financial risk categories.
- Visualize insights through an interactive Power BI dashboard.

3. Dataset Description

The dataset used in this project is a simulated financial dataset consisting of **1000 users**.

Each record represents the financial profile of a user.

Column	Description
User_ID	Unique identifier for each user
Income	Monthly income of the user
Expense	Monthly expenses
Savings	Amount saved after expenses
Debt_Payment	Monthly debt repayment
Transaction_Count	Number of financial transactions

Additional engineered features include:

Feature	Description
Savings_Ratio	Savings / Income
Debt_Burden	Debt Payment / Income
Spending_Ratio	Expense / Income
Health_Score	Combined financial stability score
Category	Financial risk classification

4. Financial Health Score Model

The financial health score is calculated using the following formula:

$$\begin{aligned} \text{Health Score} = & \\ & (\text{Savings_Ratio} \times 40) + \\ & ((1 - \text{Debt_Burden}) \times 30) + \\ & ((1 - \text{Spending_Ratio}) \times 30) \end{aligned}$$

This scoring system ensures:

- Higher savings increase financial health
- Higher debt reduces financial health

- Higher spending reduces financial health

The score ranges between **0** and **100**.

5. User Classification

Users are categorized based on their financial health score.

Score Range	Category
80 – 100	Financially Healthy
60 – 79	Moderate
40 – 59	Risky
Below 40	Vulnerable

However, in the simulated dataset used in this project, no users achieved a score above 80. Therefore, the dashboard contains three active categories: Moderate, Risky, and Vulnerable.

6. Tools and Technologies Used

Tool	Purpose
Python	Data generation and analysis
Pandas & NumPy	Data manipulation
Matplotlib & Seaborn	Exploratory data analysis
PostgreSQL	Data storage and SQL analysis
SQL	Business insights queries
Power BI	Interactive dashboard visualization

7. Data Analysis Process

The project follows a structured data analytics pipeline:

1. **Data Generation**

A synthetic dataset of financial data was generated using Python.

2. **Feature Engineering**

Financial indicators such as savings ratio, spending ratio, and debt burden were created.

3. **Financial Health Score Calculation**

A scoring model was implemented to quantify financial stability.

4. **User Classification**

Users were segmented into financial risk categories.

5. **SQL Analysis**

SQL queries were used to extract business insights.

6. **Dashboard Visualization**

Insights were visualized using Power BI.

8. Key Insights

Some major insights from the analysis include:

- **51.8% of users fall into the Risky financial category**, indicating moderate financial instability.
- **Moderate users save the highest proportion of income (~60%)** compared to other categories.
- **Vulnerable users have the highest debt burden (~18.8%)**, indicating financial risk.
- Higher income groups generally demonstrate better financial health and savings behavior.

9. Dashboard Overview

The Power BI dashboard includes the following components:

- KPI indicators for financial metrics
- Financial health score distribution

- Financial category segmentation
- Income vs savings behavior analysis
- Savings ratio analysis
- Debt burden analysis
- Income group distribution

The dashboard helps identify financial patterns and risk levels among users.

10. Conclusion

This project demonstrates how financial data analytics can be used to evaluate financial stability and identify financially vulnerable users.

By combining financial indicators into a health scoring model, financial institutions can better understand customer financial behavior and design targeted financial solutions.